# Daniel **Smullen**

*Curriculum Vitae*

*"Don't have good ideas if you aren't willing to be responsible for them." —Alan Perlis*

## About Me

I design privacy-enhancing technologies, interoperable standards, usable privacy and security controls, and AI-enabled trust systems for connected-device ecosystems, smart homes, and real-world infrastructure. My work combines software engineering, privacy engineering, human-computer interaction, requirements engineering, machine learning, data governance, Internet of Things (IoT) systems, and technical standards.
**I help organizations develop usable, secure, privacy-preserving, and trustworthy software systems.**

## Education

PhD **Software Engineering**, *Carnegie Mellon University*, Pittsburgh
Software and Societal Systems Department (formerly Institute for Software Research), Committee: Norman Sadeh (Chair), Lorrie Faith Cranor, Alessandro Acquisti, Rebecca Weiss (External, Mozilla), Yaxing Yao (External, UMBC)

- Research focused on usable privacy and security across browsers, mobile apps, and the Internet of Things, using mixed methods from HCI, behavioral economics, user-centered design, requirements engineering, machine learning, and empirical software engineering.
- Dissertation work assessed privacy and security controls for effectiveness, manageability, public policy implications, and alignment with people's mental models; it also showed how machine learning can reduce user burden while preserving preference accuracy.

MSc **Software Engineering**, *Carnegie Mellon University*, Pittsburgh, Software and Societal Systems Department

BEng **Software Engineering, Honours with Distinction**, *Ontario Tech University (formerly University of Ontario Institute of Technology)*, Oshawa

## Professional Work Experience

2024–Present **Principal Privacy Research Engineer**, *CableLabs, Security & Privacy Technologies*, Remote
I lead privacy and trust R&D for CableLabs' Security and Privacy Technologies group, specializing in usable privacy and security for connected-device ecosystems, IoT, privacy-enhancing technologies, and technical standards.

- Report to CableLabs CISO Brian Scriber while working closely with public policy, product strategy, and industry collaboration teams.
- Translate socio-technical privacy goals into architectures, protocol requirements, data taxonomies, AI-aware usability guidance, and practical engineering controls for cable-industry scale.
- Represent CableLabs in standards development and ecosystem work, including IETF Internet-Drafts and Connectivity Standards Alliance activity across Matter, Zigbee, and Aliro.
- Chair Connectivity Standards Alliance Data Privacy Working Group activity through the Marketing and Product Subgroup and the Data Export Strategic Incubator Tiger Team.
- Provide trusted privacy engineering guidance to CableLabs partners, executives, public policy teams, product strategists, and standards participants.

*https://www.daniel-smullen.com* • *daniel-r-smullen* • *drspangle*

2023–Present **Founding Ambassador and Standards Committee Member**, *Institute of Operational Privacy Design (IOPD)*, Volunteer

I serve as a volunteer Founding Ambassador and Standards Committee Member, helping develop privacy design standards that make privacy-by-design and privacy-by-default claims concrete, auditable, and implementable.

- Translate privacy principles, controls, evidence requirements, assurance criteria, and certification expectations into practical standards language for products, services, and business processes.

2023–2024 **Applied Scientist II**, *Amazon Lab126, Alexa Sensitive Content Intelligence*, Remote

I built AI-powered and LLM-based systems for large-scale Alexa customer feedback analysis, including an Alexa+ pipeline that treated chat transcripts as reading-comprehension and information-extraction tasks.

- Used off-the-shelf Large Language Models, prompting methods, and structured extraction to identify actionable trust, safety, and product-quality signals from conversational feedback.
- Avoided the cost, development time, data labeling, and maintenance burden of custom BERT-style model development and training data pipelines.
- Led empirical modeling work to assess, predict, and proactively mitigate customer-trust risks in Alexa interactions using prompt design, privacy metrics, and mixed-methods research.

2021–2023 **Applied Scientist II**, *Amazon Lab126, Devices & Services Trust and Privacy*, Remote

I worked across Amazon Devices & Services on technical privacy and trust problems that connected research, development, privacy engineering, legal decision-making, and product execution.

- Developed interpretable machine learning and decision-tree models that cross-referenced data type classifications with legal attestation decisions to predict and explain likely personally identifiable information (PII) handling.
- Served as penholder for an internal generative AI data use policy, helping translate privacy, legal, data governance, and product-risk requirements into implementable guidance for AI-enabled product teams.
- Designed privacy-preserving architectures and compliance tooling for products, processes, Privacy Enhancing Technologies (PETs), and privacy requirements engineering.
- Served as a subject matter expert for product, legal, leadership, and engineering teams on generative AI, privacy metrics, Privacy by Design, technical education, mentorship, and risk.

2015 **334F Research Associate (Radar Science and Instrument Engineering)**, *NASA Jet Propulsion Laboratory*, Pasadena

Supervisor: Razi Ahmed

- Developed a high-performance, resolution-preserving image processing pipeline for interferometric synthetic aperture radar (InSAR) imagery.
- Implemented cross-platform Python/C11 software with OpenMP multithreading and MPI multiprocessing for workstations, commodity hardware, and NASA Ames Pleiades high-performance computing.
- Improved practical processing time from weeks to minutes and contributed production radar science software for the UAVSAR program.

2012 **Engineer-in-Training (Co-op)**, *SNC-Lavalin, Global Information Technologies*, Toronto

- Assessed mission-critical datacenter systems, reliability, disaster recovery, emergency response planning, infrastructure risk, and observed system performance using Splunk and operational data.
- Developed recommendations for infrastructure improvement, risk management, workstation staging, and replacement of legacy DOS infrastructure.
- Reduced workstation deployment time from hours to minutes through improved staging and automation.

2011 **Support Technician**, *SNC-Lavalin, Mining and Metallurgy*, Toronto

- Provided dedicated technical support for high-value clients, engineering departments, and senior management.
- Developed automated data recovery workflows and performed high-risk computer forensics and recovery on damaged hardware.
- Delivered internal seminars on computer and network security, system repair, open source software, and computer forensics.

2009 **Laboratory Systems Administrator**, *DESSAU (LVM-Technisol)*, Toronto

- Managed migration and replacement of laboratory information management systems during the JEGEL/LVM-Technisol and DESSAU integration.
- Supported geotechnical field operations, subsurface sampling, concrete and asphalt QA/QC testing, and airport and highway resurfacing projects.
- Bridged laboratory information management, field data, and IT systems during active civil infrastructure work.

2009 **Laboratory Systems Administrator**, *John Emery Geotechnical Engineering Ltd.*, Toronto

- Supported City of Toronto special projects, the Toronto 2009 Capital Works Program, and Trans-Canada Highway Rehabilitation work.
- Managed geoinformatics, utility-location coordination, subsurface-sampling safety workflows, and collaborative development of ISO 9001 certified laboratory sample management software.
- Repaired legacy pavement-analysis embedded systems and designed file-storage, database, and datacenter infrastructure for the Toronto engineering office.

## Academic Work Experience

2021 **Postdoctoral Researcher**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)

Researched user-centered privacy and security controls for managing data practices embedded in browsers and web systems, extending dissertation work in usable privacy and security, HCI, and privacy engineering into applied questions about browser and web-platform control design.

2014–2021 **Graduate Research Assistant**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)

Conducted Ph.D. research in usable privacy and security using mixed-methods qualitative and quantitative methods from HCI, behavioral economics, user-centered design, requirements engineering, machine learning, and empirical software engineering. The work produced publications, datasets, and prototype systems that later became evidence and technical grounding for policy discussions, privacy-engineering practice, and standards-related work on usable privacy and security.

2017–2021 **Research Advisor**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)

Mentor for 3 undergraduate research programmers and 3 M.Sc. students.

- Employed pair programming and extreme programming techniques.
- Employed Agile software development methodologies.
- Provided practical training on test-driven development, debugging, and version control techniques.
- Provided hands-on training with research methods, including crowd-sourcing, survey design, contextual interviews, and grounded/thematic analysis on a variety of data.

2016–2020 **Teaching Assistant**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)

- 17-655/17-755: Architectures for Software Systems
- 17-781/45-887: Mobile and IoT Computing Services

2013–2014 **Research Associate**, *UOIT (now Ontario Tech University)*, Oshawa

Software Quality Research Lab, Advisor: Jeremy Bradbury

- Investigated privacy exposure in network metadata unknowingly released through ordinary Internet traffic, in an ethics board-approved user study.
- Investigated the software testing coupling effect using mutation testing tools, automated software analysis, and software-quality research methods.
- Research funded by NSERC/CRSNG (Canada).

2013 **Undergraduate Research Assistant**, *UOIT (now Ontario Tech University)*, Oshawa
Software Engineering Lab, Advisor: Ramiro Liscano

- Built TinyOS-based wireless sensor network tooling for policy programming, including Policy IDE, TOSServ, and IPv6-enabled Finger2IPv6 remote development variants for embedded systems.
- Explored how policy-based programming could make distributed embedded systems easier to develop, debug, and manage.
- Research funded by NSERC/CRSNG (Canada) Undergraduate Student Research Award.

## Projects and Funded Research

2025–Present **Privacy Preference Declaration (PPD) for Home Networks**, *CableLabs*, Remote, Co-author: Brian Scriber
Standards and privacy engineering project for user-defined privacy preferences in home networks, including IETF Internet-Drafts on protocol architecture, device accountability, signed receipts, and a machine-readable taxonomy for device data handling practices.

2025 **Defining Privacy Engineering as a Profession**, *Carnegie Mellon University*, Pittsburgh
Qualitative privacy engineering study identifying practitioner responsibilities, collaboration patterns, core competencies, hiring expectations, and professionalization challenges. Funded by: NSF CCF-2217771.

2023–2024 **Alexa+ Feedback Processing Pipeline**, *Amazon Lab126*, Remote
Internal applied machine learning project using off-the-shelf LLMs, prompt engineering, reading-comprehension framing, and structured extraction to process Alexa+ chat transcripts and customer feedback without custom BERT-style models or dedicated training datasets.

2022–2023 **Privacy Attestation PII Prediction and Explanation**, *Amazon Lab126*, Remote
Internal privacy engineering project using data type classification, legal attestation decisions, interpretable decision trees, PII prediction, and model explanations to turn privacy review patterns into actionable engineering signals.

2023 **Generative AI Data Use Policy**, *Amazon Lab126*, Remote
Internal privacy policy and data governance initiative where I served as penholder for generative AI data use guidance, helping translate privacy, legal, responsible AI, customer-trust, and product-risk requirements into implementable guidance for AI-enabled product teams.

2020–2022 **Usable Privacy and Security Controls**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Research thread on browser privacy controls, web privacy tools, Tor Browser adoption, mobile-payment security nudges, and user-centered privacy and security control design. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486, CNS-1330596, and SES-1513957.

2020–2021 **Engineering Usable Privacy and Security Affordances for Notice and Choice**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Funded research project on user-centered privacy and security affordances for notice, choice, preference management, and control interfaces. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.

2019–2020 **Design and Evaluation of Security and Privacy Nudges**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Funded research project on informational, planning, privacy, and security nudges that help users translate privacy and security intentions into action. Funded by: NSF CNS-1330596, SES-1513957, CNS-1801316.

2019 **Mobile App Privacy System (MAPS)**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Large-scale mobile app privacy compliance analysis system for crawling and analyzing the Google Play Store ecosystem. Funded by: NSF CNS-1330596, CNS-1330141, CNS-1330214; XSEDE ACI-1548562.

2017–2019 **IoT Resource Registry**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Personalized Privacy Assistant and IoT Privacy Infrastructure project for machine-readable service descriptions, contextual notice, choice, and privacy preference management in connected environments. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.

2018–2019 **Internet of Things Privacy Infrastructure (Patent Pending)**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Funded research project on IoT privacy infrastructure, resource registries, service descriptions, and mechanisms for contextual notice and choice. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.

2017–2019 **DARPA Brandeis: A Privacy Assistant for the Internet of Things**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Funded research project developing privacy assistant infrastructure for nearby IoT services, sensors, smart spaces, and context-aware systems. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.

2017–2019 **The Usable Privacy Policy Project**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Privacy policy analysis project combining crowdsourcing, natural language processing, machine learning, privacy policy annotations, mobile app privacy compliance, and usable privacy policy research. Funded by: NSF CNS-1330596, CNS-1330141, CNS-1330214.

2017 **Mobile App Privacy Compliance System (Patent Pending)**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Funded research project underlying MAPS, focused on scalable mobile app privacy compliance analysis across code-level practices and privacy policy disclosures. Funded by: NSF CNS-1330596, CNS-1330141, CNS-1330214; XSEDE ACI-1548562.

2016 **Eddy: A Privacy Requirements Specification Language**, *Carnegie Mellon University*, Pittsburgh, Advisor: Travis Breaux
Privacy requirements engineering language and formal analysis tool for collection, use, transfer, retention, compliance reasoning, and consistency checking over privacy obligations. Funded by: NSF CNS-1330596; ONR N002441410028; National Security Agency.

2015 **Non-Local Interferometric SAR Parameter Estimator**, *NASA Jet Propulsion Laboratory*, Pasadena, Advisor: Razi Ahmed
High-performance, resolution-preserving InSAR filtering system for UAVSAR imagery, using cross-platform Python/C11, OpenMP, MPI, and supercomputing deployment to reduce processing time from weeks to minutes. Funded by: NASA JPL PhD Fellowship.

2015 **NL-InSAR with Gaussian-Laplacian Pyramids**, *NASA Jet Propulsion Laboratory*, Pasadena, Advisor: Razi Ahmed
Radar image processing project extending non-local InSAR filtering with Gaussian-Laplacian pyramids to reduce canvas effects while preserving reflectivity, interferometric phase, coherence, and fine image structure.

2014 **Incident Recognition and Intelligence System (IRIS)**, *Ontario Tech University*, Oshawa, Advisor: Shahryar Rahnamayan
Computer vision, web, and traffic-monitoring system for highway incident recognition, video review, statistical aggregation, and situational awareness. Funded by: University of Toronto ITS Laboratory, Ontario Tech University, NSERC/CRSNG.

2014 **Automated Marking System (AMS)**, *Ontario Tech University*, Oshawa, Advisor: Kamran Sartipi
Ruby on Rails educational technology system for creating, distributing, compiling, executing, and automatically grading introductory programming assignments in a protected environment. Funded by: Ontario Tech University.

2013 **Policy IDE, TOSServ, Finger2IPv6**, *Ontario Tech University*, Oshawa, Advisor: Ramiro Liscano

TinyOS wireless sensor network tooling for policy-based programming, debugging policy-driven embedded applications, server-side TinyOS workflows, and IPv6-enabled remote development. Funded by: NSERC/CRSNG Undergraduate Student Research Award.

2013 **Military Logistics Management System (MLMS)**, *Ontario Tech University*, Oshawa, Advisor: Eyhab Al-Masri

ASP.NET military logistics dashboard and visualization system for supply, personnel, operational reporting, role-aware access, and command-and-control style situational awareness. Funded by: Ontario Tech University.

2012 **sneakyFS File System Design and Implementation**, *Ontario Tech University*, Oshawa, Advisor: Kamran Sartipi

UNIX-style C file system on a simulated hard disk with indexed and linked allocation, transparent encryption, journaling, UUIDs, low-level memory management, and secure storage design. Funded by: Ontario Tech University.

2012 **Datacenter Utilization Research Study**, *SNC-Lavalin Global Information Technologies*, Toronto, Advisor: Marc Ross

Mission-critical datacenter utilization, availability, performance, disaster recovery, and infrastructure remediation study for critical server assets at the West Mall campus datacenter. Funded by: SNC-Lavalin.

2009 **Laboratory Information Management System (LIMS)**, *John Emery Geotechnical Engineering Ltd.*, Toronto, Advisor: John Emery

ISO 9000-certified laboratory information management software for geotechnical and materials-testing workflows, including sample intake, process queues, laboratory status tracking, and production reporting.

2009 **Trans-Canada Highway Rehabilitation Geoinformatics**, *John Emery Geotechnical Engineering Ltd.*, Toronto, Advisor: Alain Duclos

Geoinformatics and civil engineering support project for large-scale asphalt rehabilitation planning across more than 300 km of the Trans-Canada Highway near Banff, including technical drawings, probe-hole data, pavement conditions, and remediation planning.

## Refereed Publications

**D. Smullen**, "Privacy Preference Declarations: Signed receipts for expectation-aligned smart homes," in *2025 IEEE Symposium on Privacy Expectations (ISoPE)*. IEEE, 2025, pp. 14–17.

N. Samarin, N. R. Narla, L. Webster, and **D. Smullen**, "Defining privacy engineering as a profession," *Proceedings on Privacy Enhancing Technologies*, vol. 2025, no. 4, pp. 549–565, 2025.

P. Story, **D. Smullen**, R. Chen, Y. Yao, A. Acquisti, L. F. Cranor, N. Sadeh, and F. Schaub, "Increasing adoption of Tor Browser using informational and planning nudges," *Proceedings on Privacy Enhancing Technologies*, vol. 2022, no. 2, pp. 152–183, 2022.

**D. Smullen**, "Informing the design and refinement of privacy and security controls," PhD dissertation, Carnegie Mellon University, Sep. 2021.

P. Story, **D. Smullen**, Y. Yao, A. Acquisti, L. F. Cranor, N. Sadeh, and F. Schaub, "Awareness, adoption, and misconceptions of web privacy tools," *Proceedings on Privacy Enhancing Technologies*, vol. 2021, no. 3, pp. 308–333, 2021.

**D. Smullen**, Y. Yao, Y. Feng, N. Sadeh, A. Edelstein, and R. Weiss, "Managing potentially intrusive practices in the browser: A user-centered perspective," *Proceedings on Privacy Enhancing Technologies*, vol. 2021, no. 4, pp. 500–527, 2021.

P. Story, **D. Smullen**, A. Acquisti, L. F. Cranor, N. Sadeh, and F. Schaub, "From intent to action: Nudging users

towards secure mobile payments," in *Sixteenth Symposium on Usable Privacy and Security (SOUPS 2020)*, 2020, pp. 379–415. [Online]. Available: https://www.usenix.org/conference/soups2020/presentation/story

**D. Smullen**, Y. Feng, S. A. Zhang, and N. Sadeh, "The best of both worlds: Mitigating trade-offs between accuracy and user burden in capturing mobile app privacy preferences," *Proceedings on Privacy Enhancing Technologies*, vol. 2020, no. 1, pp. 195–215, 2020.

S. Zimmeck, P. Story, **D. Smullen**, A. Ravichander, Z. Wang, J. Reidenberg, N. C. Russell, and N. Sadeh, "MAPS: Scaling privacy compliance analysis to a million apps," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 3, pp. 66–86, 2019.

P. Story, S. Zimmeck, A. Ravichander, **D. Smullen**, Z. Wang, J. Reidenberg, N. C. Russell, and N. Sadeh, "Natural language processing for mobile app privacy compliance," in *AAAI Spring Symposium on Privacy-Enhancing Artificial Intelligence and Language Technologies*, 2019. [Online]. Available: https://ceur-ws.org/Vol-2335/1st_PAL_paper_6.pdf

S. Wilson, F. Schaub, F. Liu, K. M. Sathyendra, **D. Smullen**, S. Zimmeck, R. Ramanath, P. Story, F. Liu, N. Sadeh, and N. A. Smith, "Analyzing privacy policies at scale: From crowdsourcing to automated annotations," *ACM Transactions on the Web*, vol. 13, no. 1, Dec. 2018.

A. Das, M. Degeling, **D. Smullen**, and N. Sadeh, "Personalized privacy assistants for the internet of things: Providing users with notice and choice," *IEEE Pervasive Computing*, vol. 17, no. 3, pp. 35–46, 2018.

**D. Smullen** and T. Breaux, "Improving security in software acquisition with data retention specifications," US Navy Acquisition Research Program Sponsored Report Series, Tech. Rep., 2017. [Online]. Available: https://calhoun.nps.edu/handle/10945/58892

J. Bhatia, T. Breaux, L. Friedberg, H. Hibshi, and **D. Smullen**, "Privacy risk in cybersecurity data sharing," in *ACM Workshop on Information Sharing and Collaborative Security (WISCS)*, 2016, pp. 57–64.

**D. Smullen** and T. Breaux, "Modeling, analyzing, and consistency checking privacy requirements using Eddy," in *Proceedings of the Symposium and Bootcamp on the Science of Security*, 2016, pp. 118–120.

T. Breaux, **D. Smullen**, and H. Hibshi, "Detecting repurposing and over-collection in multi-party privacy requirements specifications," in *IEEE International Requirements Engineering Conference*, 2015, pp. 166–175.

**D. Smullen** and T. Breaux, "Toward rapid recertification using formal analysis," US Navy Acquisition Research Program Sponsored Report Series, Tech. Rep., 2015. [Online]. Available: https://www.acquisitionresearch.net/publications/detail/1532/

**D. Smullen**, J. Gillett, J. Heron, and S. Rahnamayan, "Genetic algorithm with self-adaptive mutation controlled by chromosome similarity," in *IEEE Congress on Evolutionary Computation (CEC)*, 2014, pp. 504–511.

N. Qwasmi, **D. Smullen**, and R. Liscano, "Integrated development environment for debugging policy-based applications in wireless sensor networks," *Procedia Computer Science*, vol. 21, pp. 225–233, 2013.

## Standards and Technical Drafts

**Privacy Preference Declaration for Home Networks**, *IETF Internet-Draft, work in progress*. Authored with CableLabs CISO Brian Scriber. Defines a framework for user-defined privacy preferences across home networks, including policy discovery, device accountability, and privacy governance for connected-device ecosystems.

**Privacy Preference Declaration Taxonomy**, *IETF Internet-Draft, work in progress*. Authored with CableLabs CISO Brian Scriber. Defines a machine-readable taxonomy for describing device data handling practices, including data types, purposes, actions, sources, and destinations.

## Talks, Posters, and Recorded Sessions

2026 **PETs in Practice: From Privacy Policies to User-Held Preferences with CableLabs**, *Duality Technologies PETs in Practice*, Online, Recorded Series
- Discussed privacy-enhancing technologies, user-held preferences, and practical deployment paths for privacy engineering in connected-device ecosystems.

2025 **AI's Best Friend: The Role of PETs, Enabling Data Use & Operationalizing Privacy**, *IAPP Privacy. Security. Risk. 2025*, San Diego, Panel
- Participated in an expert panel on Privacy Enhancing Technologies, privacy operations, responsible data use, and privacy engineering practice.

2025 **Privacy Preference Declarations: Signed Receipts for Expectation-Aligned Smart Homes**, *IEEE Symposium on Privacy Expectations (ISoPE 2025)*, New York City, Paper Session
- Presented conference paper on Privacy Preference Declarations, signed receipts, smart-home expectations, and user-centered privacy governance for connected devices.

2023 **Data-Driven Methodologies Toward More Usable Personalized Settings**, *Amazon DataCon*, Online, Conference Talk
- Presented an award-winning vision for the future of usable privacy and security as a featured subject matter expert, showcasing machine learning and mixed-methods research techniques.

2022 **Information Privacy Engineering**, *University of Maine School of Computing and Information Science*, Online, Guest Lecture
- Provided lectures and discussion session as a featured subject matter expert in a graduate level academic course.

2020 **The Best of Both Worlds: Mitigating Trade-offs Between Accuracy and User Burden in Capturing Mobile App Privacy Preferences**, *20th Privacy Enhancing Technologies Symposium (PETS 2020)*, Online, Conference Talk
- Presented conference talk for peer reviewed publication and participated in panel discussions.

2019 **Digital Data Flows Masterclass: Mobile Apps**, *Future of Privacy Forum*, Online, Masterclass
- Provided lectures and discussion session as a featured subject matter expert.

2018 **Personalized Privacy Assistant Project**, *US Federal Trade Commission PrivacyCon 2018*, Washington D.C., Research Poster
- Presented poster with oral presentation and participated in panel discussions.

2017 **Assisting Users in a World Full of Cameras: A Privacy-aware Infrastructure for Computer Vision Applications**, *US Federal Trade Commission PrivacyCon 2017*, Washington D.C., Research Poster
- Presented poster with oral presentation and participated in panel discussions.

2017 **A Privacy Assistant for the Internet of Things**, *13th Symposium On Usable Privacy and Security (SOUPS 2017)*, San Jose, Research Poster
- Presented poster with oral presentation and participated in panel discussions.

2016 **Toward a Semantics for Data Retention in Eddy**, *National Institute of Standards and Technology (NIST)*, Gaithersburg, Invited Talk
- Delivered an invited research talk, hosted by NIST Applied Cybersecurity Division.

2015 **Privacy Engineering Tool Clinic**, *Computing Community Consortium Catalyst, Privacy by Design Workshop*, Pittsburgh, Invited Talk and Tool Demonstration
- Delivered an invited talk with tool demonstration and discussion panel session.

2015 **Detecting Repurposing and Over-Collection in Multi-party Privacy Requirements Specifications**, *23rd IEEE International Requirements Engineering Conference (RE 2015)*, Ottawa, Conference Talk
- Presented conference talk for peer reviewed publication and participated in panel discussions.

2015 **Towards Rapid Re-Certification Using Formal Analysis**, *US Navy Acquisition Research Symposium*, Monterey, Conference Talk

- Presented conference talk for peer reviewed publication and participated in panel discussions.

2014 **Genetic Algorithm with Self-Adaptive Mutation Controlled by Chromosome Similarity**, *IEEE World Congress on Computational Intelligence (WCCI 2014), Evolutionary Computation Conference (CEC 2014)*, Beijing, Conference Talk

- Presented conference talk for peer reviewed publication and participated in panel discussions.

2013 **How Much Do We Reveal Through Metadata? An Assessment of Online Privacy**, *IBM Consortium for Software Engineering Research (CSER 2013)*, Toronto, Preliminary Study Poster

- Presented poster with oral presentation and participated in panel discussions.

2013 **Policy IDE... and Lessons Learned Since**, *The 4th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN-2013)*, Niagara Falls, Conference Talk

- Presented conference talk for peer reviewed publication and participated in panel discussions.

2013 **Facilitating the Internet of Things with Policy Programming**, *UOIT (now Ontario Tech University) Undergraduate Research Showcase 2013*, Oshawa, Research Poster

- Presented poster with oral presentation and participated in panel discussions.

## Professional Service

2026-Present **Program Committee Member**, *IEEE Symposium on Privacy Expectations (SoPE)*, Conference

- Review submissions and contribute to program committee discussions for the IEEE symposium on privacy expectations.

2020-Present **External Program Committee Member**, *Conference on Human Factors in Computing Systems (ACM SIGCHI)*, Conference

- Reviewed submissions in the main proceedings and Late Breaking Works.
- Shepherded papers in the main proceedings and Late Breaking Works tracks.

2021-Present **Reviewer**, *IEEE Access*, Online, Journal

- Reviewed submissions and participated in editorial discussions.

2021-2022 **Reviewer**, *ACM Transactions on Privacy and Security (TOPS, formerly known as TISSEC)*, Online, Journal

- Reviewed submissions and participated in editorial discussions.

2021 **Program Chair**, *ACM SIGBOVIK*, Online, Conference

- Organized conference and webcast, acting as moderation chair for Q&A.

2020-2021 **Council Member**, *Cylab Justice Equity Diversity and Inclusion (JEDI) Council*, Carnegie Mellon University

- Helped establish a new committee to promote justice, equity, diversity, and inclusion in faculty candidates, talks, hiring and student affairs.

2019-2020 **Committee Member**, *Institute for Software Research Re-Branding Committee*, Carnegie Mellon University

- Participated in coordinating and analyzing the results of a market research campaign for re-branding the department, establishing a new messaging/communications strategy, and developing a novel recruiting strategy for diverse faculty and doctoral students. The department was re-branded successfully in 2022, and is now called the Software and Societal Systems Department.

2018-2021 **Co-Founder**, *Software and Societal Systems Department Forge Makerspace*, Carnegie Mellon University

- Performed layout, development, maintenance, and instruction for the makerspace, leading safety training and other projects related to developing CNC machining, 3D printing, and CAD capabilities for the department.

2018-2020 **Co-Founder**, *Software and Societal Systems Department Lunch and Learn Seminar Series*, Carnegie Mellon University, Seminar

- Managed and taught weekly seminars concerning a variety of software engineering topics, aimed at improving software development best practices among researchers.

2018 **Program Committee Member**, *European Conference on Information Systems (ECIS 2018)*, Portsmouth, Conference

- Reviewed submissions and participated in program committee discussions and shepherding.

2016 **Program Committee Member**, *International Workshop on Privacy Engineering (IWPE 2016)*, San Jose

- Reviewed submissions and participated in program committee discussions and shepherding.

2015 **Web Chair**, *International Conference on Multicore Software Engineering, Performance, and Tools (MUSEPAT 2014)*, Hong Kong, Conference

- Organized conference website and coordinated program committee.

2015 **Review Committee Member**, *Software and Societal Systems Department PhD Applicant Review Committee*, Carnegie Mellon University

- Reviewed applications for incoming doctoral students and participated in the final acceptance decision panel.

## Awards

2024 **Outstanding Contributor Award**, *Connectivity Standards Alliance Data Privacy Working Group*

2017 **Distinguished Poster Award**, *13th Symposium On Usable Privacy and Security (SOUPS 2017)*, San Jose

2017 **Hima and Jive Fellowship in Computer Science for International Students**, *Carnegie Mellon University*, Pittsburgh, Pennsylvania

2015 **Ready-Set-Transfer Award**, *23rd IEEE International Requirements Engineering Conference*, Ottawa, *"Eddy: A privacy requirements specification language"*

- Awarded first place in competitive industrial panel talks.

2014 **Faculty of Engineering and Applied Science Undergraduate Capstone Design Challenge Winner**, *Ontario Tech University*, Oshawa, *"Incident Recognition and Intelligence System (IRIS)"*

- Awarded first place in capstone research project competition, evaluated by a panel of academic and industrial experts.

2012–2014 **President's Honours List**, *Ontario Tech University*, Oshawa

- Awarded for exceptional academic achievement, with greater than 3.7 GPA.

2013 **National Science and Engineering Research Council (NSERC/CRSNG) Undergraduate Research Award**, *Ontario Tech University*, Oshawa

- Awarded federally funded research grant with competitive review process, funding 1 year of undergraduate research in engineering.

2011 **Dean's Honours List**, *Ontario Tech University*, Oshawa

- Awarded for exceptional academic achievement, with greater than 3.5 GPA.

2008 **Engineers Without Borders Design Challenge Winner**, *McMaster University*, Hamilton
- Awarded first place in a software requirements specification competition for distribution of AIDS medication in rural Africa.

2007 **DaVinci Engineering Design Challenge Winner**, *University of Toronto*, Toronto
- Awarded first place in an engineering design challenge for remote controlled electric motorized aquatic vehicles, in Electrical Engineering and Fluid Dynamics competition stream.

## Languages and Technologies

English **Native Proficiency**

French **Verbal, Written Proficiency**

German **Verbal, Written Proficiency**

Python **Preferred Language**, *My default programming language, used for everything from web applications with django and django-rest, to data analysis and machine learning with scikit-learn, to performance analysis and scalability optimization with Cython in combination with several different multiprocessing and multithreading libraries*

Docker **Preferred Technology**, *My approach is to containerize as much as possible, employing a DevOps approach to research projects, prototypes, and larger-scale projects alike. Docker makes managing dependencies quicker and easier, and streamlines deployment across different platforms and architectures*

R **Preferred Language**, *Used in several research projects for statistical analysis, regression modelling, data mining, exploration and visualization*

JavaScript **Secondary Preference**, *Used in the development of full-stack web applications using MongoDB, Express Angular, and Node, geo-spatial mapping libraries, and cross-platform hybrid mobile applications using a variety of technologies such as Ionic*

C++ **Secondary Preference**, *Used in high performance computing work for optimized image processing kernels and computer vision applications, a useful alternative when performance is a primary architectural driver*

Java **Secondary Preference**, *Used to develop parsers, lexers, and compilers for domain specific languages such as Eddy, as well as interact with a variety of Java-based research tools such as OWL-DL and other logic engines*

## Professional Memberships

Since 2024 **Connectivity Standards Alliance (the Alliance)**, *Contributing Member*
- Data Privacy Working Group
- Data Export Strategic Incubator Tiger Team, Chair
- Matter Working Group
- Zigbee Working Group
- Aliro Working Group
- Certification Subgroup
- Technical Subgroup
- Marketing and Product Subgroup, Chair

Since 2023 **Institute of Operational Privacy Design (IOPD)**, *Founding Ambassador, Standards Committee Member*

Since 2023 **Association for Computing Machinery (ACM)**, *Professional Member*

Since 2021 **International Association of Privacy Professionals (IAPP)**, *Member*

Since 2020 **HACK Pittsburgh**, *Member and Educator*

Since 2018 **Software and Societal Systems Department Forge Makerspace**, *Founding Member*, Carnegie Mellon University

Since 2014 **Collaborative Institutional Training Initiative**, *Responsible Conduct of Research (Physical Science, Social & Behavioral Research)*, Carnegie Mellon University

Since 2014 **The Corporation of the Seven Wardens**, *Order of the Calling of the Engineer*, Camp 1, Toronto, Ontario, Canada

Since 2013 **Institute of Electrical and Electronics Engineers (IEEE)**, *Professional Member*, 92797682

- IEEE Computer Society Technical Community on Pattern Analysis and Machine Intelligence
- IEEE Computer Society Technical Community on Security and Privacy
- IEEE Internet of Things Community
- IEEE Digital Privacy Community
- IEEE TechEthics Community

Since 2011 **International Red Cross**, *Class C Emergency First Aid, CPR and Defibrillator Certification*